Subtractive Mixture Models via Squaring: Representation and Learning [2]

Paper Code

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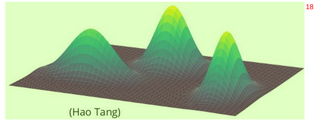
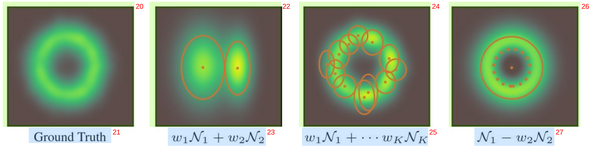
0 Mixture models [16]

$$p(\mathbf{X}) = \sum_{i=1}^{K} w_i \, p_i(\mathbf{X}) \quad \text{subject to} \quad w_i \geq 0 \quad \sum_{i=1}^{K} w_i = 1$$ [17]


(Hao Tang) [18]

✗ components can only be added together! [19]



Fewer components with subtractions [28]

Questions? [29] **...Contributions!** [30] 1 2 3 [31]

1 How to learn subtractive mixture models? [32]

$$p(\mathbf{X}) = \sum_{i=1}^{K} w_i \, p_i(\mathbf{X}) \qquad w_i \in \mathbb{R}$$ [33]

How to ensure $p(\mathbf{X})$ is non-negative? [34]

$\Longrightarrow$ Impose ad-hoc constraints over the parameters [35]

✗ challenging to derive in closed-form [1][2][3] [36]

2 How much more expressive are they? [37]

with respect to traditional additive-only mixtures [38]

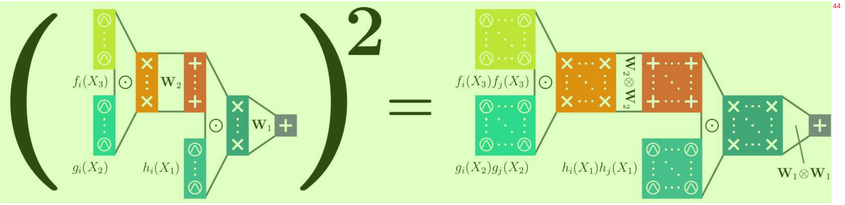
3 What is their relationship with other models? [39]

understanding why they are expressive ... [40]

... and why they support tractable inference [41]

TL; DR *"We learn exponentially more expressive mixture models with subtractions, by squaring deep tensorized mixtures"* [42]

Learning deep subtractive mixtures by squaring layers of a deep circuit [43]

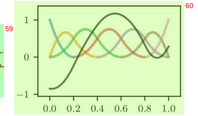

 [44]

1 Squaring mixtures ... [56]

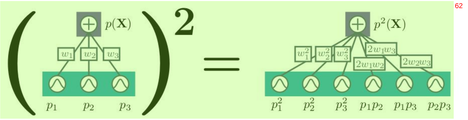
$$p(\mathbf{X}) \propto \left(\sum_{i=1}^{K} w_i \, p_i(\mathbf{X}) \right)^2 = \sum_{i=1}^{K} \sum_{j=1}^{K} w_i w_j \, p_i(\mathbf{X}) p_j(\mathbf{X})$$ [57]

Renormalization: [58]

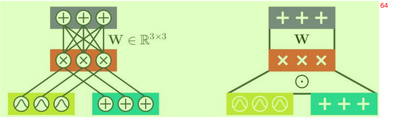
$$Z = \sum_{i=1}^{K} \sum_{j=1}^{K} w_i w_j \int p_i(\mathbf{X}) p_j(\mathbf{X}) \mathrm{d}\mathbf{X}$$ [59]


 [60]

Tractable marginalization is supported by exponential families [2] and splines components [61]


 [62]

... by squaring circuits [63]

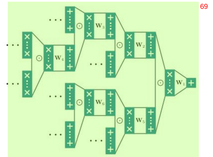

 [64]

Build deep mixtures with layers as "Lego blocks" [65]

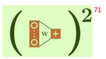
Theorem. exponential separation [4] [5] [66]

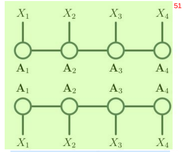
2 There is a class of distributions $\mathcal{F}$ over variables $\mathbf{X}$ that can be compactly represented as a shallow squared mixture with negative weights, but the smallest structured decomposable additive-only mixture of any depth computing any $F \in \mathcal{F}$ has size $2^{\Omega(|\mathbf{X}|)}$. [67]

Deep additive-only mixtures [68]


 [69]

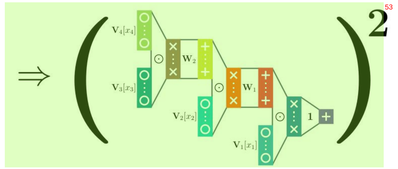
Squared subtractive mixture model [70]


 [71]



Born machines [6] [52]

$\Rightarrow$

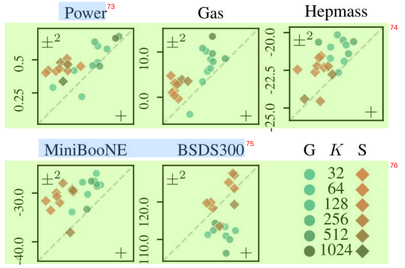

 [53]

$$p(\mathbf{x}) \propto \kappa(\mathbf{x})^\top \mathbf{A} \, \kappa(\mathbf{x})$$

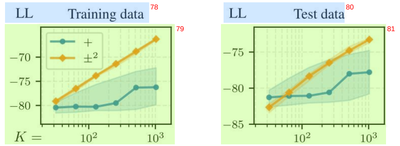
PSD models [7] $\Rightarrow \sum \left(\mathbf{w} \right)^2$ [54]

3 Understanding the expressiveness of other models in a unifying framework [55]

Density estimation [72]


 [76]

GPT2 distillation [77] [8]


 [79] [81]

References [82]

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